

Hydrogen Reciprocating Gas-Engine OEM Survey

MYSGH2 Project — P1 (H₂-fired) Singapore-offtake power block | Engine-only (turbines excluded) |
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Scope: hydrogen-capable **reciprocating** gas engines (gensets) only — gas turbines are excluded and covered separately in the project's turbine-side OEM survey. This report supports evaluation of the **P1 hydrogen-fired** power-generation option on the Singapore offtake side of the MYSGH2 ammonia-to-hydrogen value chain. Per the repository's No-Fabrication Rule, every figure below carries a cited public source; where a figure could not be found, that is stated explicitly rather than estimated.

Headline finding

No OEM publishes an electrical efficiency for a 100%-hydrogen reciprocating engine at MW scale. Wärtsilä states outright for its 31H2 that the exact output/efficiency impact of switching from natural gas to pure hydrogen “will be communicated in conjunction with the sales release” — an OEM-confirmed non-publication, not a gap in this search. Every MW-class efficiency figure in the table below is therefore quoted on a **natural-gas basis** for the same platform and must not be read as an H₂ efficiency. The only H₂-specific efficiencies located anywhere are 2G Energy's sub-MW H2 CHP models (37.7–41.2%).

Maturity split

Genuine **commercial** product today = partial **blends (25–30 vol% H₂)** on MW-class platforms. **100% H₂** firing is **demonstrator / first-of-a-kind** status at MW scale, and commercially available only in the **sub-MW** class (INNIO Jenbacher Type 4, 2G Energy).

Singapore / Southeast Asia relevance

No hydrogen reciprocating-engine deployment or certification was located in Singapore or anywhere in SEA. Every Singapore hydrogen power project identified is a **gas turbine** (Keppel Sakra Cogen 600 MW/30% H₂, H1 2026; GE Vernova 9HA.01 at Pulau Seraya; Sembcorp/MHI; Senoko Energy–Gentari feasibility on importing H₂ from Malaysia, supply from ~2029). The nearest engine reference for Asia-Pacific is INNIO's Hyosung Ulsan (South Korea) plant — the first 100%-H₂ reciprocating-engine power plant in APAC.

OEM Comparison Table

H₂ fractions are quoted as each source states them (vol% unless noted). Efficiency is natural-gas-basis electrical efficiency unless explicitly marked H₂-basis.

OEM	Model(s)	Unit size	H ₂ capability	Readiness status	Electrical efficiency	Key note
Wärtsilä	31H2 (Wartsila 31 platform)	4.6–10.4 MW	100% H ₂	Commercial — tendering 2025, delivery 2026; 100% H ₂ to Spanish grid, Bermeo, Jun 2026	Not published on H ₂ (NG sibling 31SG >50%, OEM's own claim)	World's first large-scale 100% H ₂ engine feeding a national grid
Kawasaki	KG-18-T	7.5 MW ("8 MW class" in verification framing)	30 vol% H ₂ co-firing	Commercial — sales launched 30 Sep 2025 (OEM claims world first for large gas engine)	Not obtained (source proxy-blocked)	>240 KG-series units sold since 2011
Bergen Engines	B36:45V	6,750–11,785 kW (engine) / 6,900–9,600 kW (genset)	25% H ₂ unmodified across full NG range; 100% H ₂ fired Dec 2024	Mixed — 25% commercial-ready; 100% is a demonstrator, industrialisation ongoing	Not published	Source conflict: commercial target elsewhere cited as "up to 60%, 100% for new engines to come" — dates not reconciled
Caterpillar (Cat Electric Power)	CG132B, CG170B, CG260, G3500H, G3516	400 kW–4.5 MW (blend line); G3516 demonstrator to 1,000 kW	Up to 25 vol% H ₂ (factory/retrofit); G3516 100% H ₂ demonstrator since 2022	Commercial (blends) / demonstrator (100%)	Not published on H ₂	OEM also claims 35 yrs experience with application-specific blends up to 80% H ₂ (not a product rating)
Caterpillar / MWM (Cat Energy Solutions)	TCG 3020 (V12/16/20), TCG 3016	1,380–2,300 kW (TCG 3020)	25 vol% H ₂ admixture via 25H2-Kit (new + retrofit)	Commercial (25% blend); no 100% H ₂ MWM product found	Up to 45% (NG) / 43.6% (biogas)	—
INNIO Jenbacher	Type 4 / J420 (incl. J420 D/E)	Up to 1,694 kW (J420); J420 D/E up to 1,562 kW	Type 4: 100% H ₂ available; rest of line "Ready for H2" at 25 vol%, convertible	Commercial — field refs: 1.4 MW 100% H ₂ at Hychico Diadema (Argentina); 3 MW 100% H ₂ data-centre backup demo Apr 2026; first 100% H ₂ engine plant in APAC (Hyosung, Ulsan, Korea)	Up to 42.6% (NG basis)	Closest engine reference to Asia-Pacific / Singapore of any OEM in this survey
Rolls-Royce Power Systems (mtu)	Series 4000 FNER/FV; new Series 4000 L64	L64: 2.8 MW full output from 2026	TÜV SÜD H2-ready certified 11 Jul 2024: 25 vol% blend and 100 vol%, newbuild + retrofit	Commercial cert.; first 100% H ₂ install at Enerport II (Duisburg)	Not published on H ₂	L64 fast-start: 45 s to full load
2G Energy	agenitor 404c H2, 406 H2, 408 H2, 412 H2; avus 1000plus H2	115–750 kW (sub-MW)	100% H ₂ (dedicated H2-suffix models)	Commercial — first fully H ₂ -operated CHP commissioned 2014	37.7% / 39.0% / 40.2% / 40.5% / 41.2% (H ₂ -basis — only OEM publishing this)	NG siblings (agenitor 406/408) reach 42.5% ⇒ indicative ~2.3–3.5 pt H ₂ penalty; not transferable to MW-class
DEUTZ	TCG 7.8 H2	200 kW (genset to 170 kVA electrical)	100% H ₂	Commercial — volume production from 2024; ~100 H ₂ gensets commissioned in China	Not published	Sub-MW class
HD Hyundai Infracore	HX12 (11 L)	200 kW test unit	100% H ₂ (engine programme)	Pre-commercial — 1,500 h completed on 200 kW generator; power-gen mass production targeted 2026	Not published	Trucks-first programme; power-gen is a stated future target

OEM	Model(s)	Unit size	H ₂ capability	Readiness status	Electrical efficiency	Key note
Everllence (ex-MAN ES)	35/44GTS, 51/60G; 35/44DF H2 research engine	Not stated for H ₂ research engine	35/44GTS & 51/60G: 25% H ₂ blend; 35/44DF H2 single-cylinder research engine run on 100% H ₂ (commissioned Augsburg, late 2025)	R&D; — no commercial 100% H ₂ power-gen product yet	Not published	—
Cummins	B6.7H (15 L in development)	290 hp (B6.7H, on-/off-highway)	100% H ₂ (on-/off-highway ICE)	On-/off-highway only — no power-gen genset product located (unconfirmed, not absent)	Not applicable	Out of scope pending further product confirmation
Liebherr	H966	13 L, 6-cyl	100% H ₂ (PFI)	Off-highway/construction prototype — no genset product located	Not published	Listed only so it is not mis-scoped as a power-gen offering
Yanmar	Pilot-ignition H ₂ 4-stroke (unnamed model)	~500 kW	H ₂ ratio not stated in release	Demonstrated at rated output, 30 Oct 2024	Not published	Marine-auxiliary framing, not grid power-gen

■ Conflicts flagged, not resolved: Bergen's H₂ rating is quoted as 10% / 25% / "commercial target 60%" / 100%-fired across different sources and dates — do not merge. Kawasaki's KG-18-T is stated as 7.5 MW in one release and "8 MW class" in the verification framing.

Cross-read vs. the ammonia-fired (P2) engine option

The hydrogen reciprocating engine is roughly **two product generations ahead** of the ammonia-fired engine at MW scale: a 4.6–10.4 MW machine (Wärtsilä 31H2) has fed 100% H₂ into a national grid (Jun 2026), whereas the largest ammonia-fired engines with confirmed sales dates are 1.9–3.1 MW (Wärtsilä 25 Ammonia) and 2–6 MW (IHI 28ADF, from 2027). This strengthens the case for P1 (hydrogen) on the **engine** readiness axis, independently of the turbine-axis argument already recorded in project memory (2026-07-28).

Sourced unit-size spine (largest to smallest, MW class)

- Wärtsilä 31H2 — 4.6–10.4 MW (100% H₂, grid demo Jun 2026, delivery from 2026)
- Bergen B36:45V — 6.75–11.8 MW (25% today, 100% fired Dec 2024)
- Kawasaki KG-18-T — 7.5 MW (30% H₂, commercially on sale 30 Sep 2025)
- MWM TCG 3020 — 1.38–2.3 MW (25% H₂)
- INNIO J420 — ≤1.69 MW (100% H₂ at Type 4)
- Rolls-Royce mtu 4000 L64 — 2.8 MW (25%/100% certified)
- Cat G3516 — 1.0 MW (100% H₂ demonstrator)
- 2G / DEUTZ / HD Hyundai / Yanmar — 0.2–0.75 MW (sub-MW class)

Data-quality note

Several source pages were proxy-blocked during this research pass (innio.com, 2-g.com, global.kawasaki.com, newatlas.com, h2-tech.com); figures for these were taken from search-engine extracts of the same canonical pages, with URLs still cited below. Re-open at source before any investment-grade use. No figure in this report originates from model memory alone — every quantitative claim is either sourced, labelled as an assumption, or shown as a calculation, per the repository's No-Fabrication Rule.

Sources

OEM / Org	Item	URL
Wärtsilä	31H2 world-first 100% H ₂ grid engine (Bermeo, 11 Jun 2026)	https://www.wartsila.com/media/news/11-06-2026-world-s-first-large-scale-100-hydrogen-engine-tested-at-wartsila-s-bermeo-laboratory-to-support-the-spanish-grid-3760292
Wärtsilä	Hydrogen-ready power plant solutions (efficiency impact TBC at sales release)	https://www.wartsila.com/energy/sustainable-fuels/hydrogen-power-plant
Wärtsilä	31SG gas engine, >50% simple-cycle efficiency claim	https://www.wartsila.com/energy/engine-power-plant-solutions/engine-power-plant-products/wartsila-31sg-gas-engine
Kawasaki	KG-18-T commercial launch, 30% H ₂ co-firing (30 Sep 2025)	https://global.kawasaki.com/en/corp/newsroom/news/detail/?f=20250930_4865
Kawasaki	Green Gas Engine line-up	https://global.kawasaki.com/en/energy/equipment/gas_engines/line_up.html
Bergen Engines	B36:45V genset datasheet (6,900–9,600 kW)	https://www.bergenengines.com/wp-content/uploads/2024/11/Bergen-B36-45V-Genset.pdf
Bergen Engines	B36:45V engine datasheet (6,750–11,785 kW)	https://www.bergenengines.com/wp-content/uploads/2021/12/Bergen-Engines-B36-45V.pdf
Modern Power Systems	Bergen demonstrates 100% hydrogen (incl. ≤60% commercial-target wording)	https://www.modernpowersystems.com/analysis/bergen-demonstrates-100-hydrogen/
Caterpillar	CG260 added to hydrogen lineup (25 vol% blend)	https://www.cat.com/en_US/news/engine-press-releases/caterpillar-advances-hydrogen-program-by-adding-cat-cg260-gas-generator-set-to-lineup.html
Caterpillar	400 kW–4.5 MW hydrogen-blend portfolio + G3516 100% H ₂ demonstrator (1,000 kW)	https://www.cat.com/en_US/news/engine-press-releases/caterpillar-expands-lineup-of-hydrogen-fueled-power-solutions-with-generator-sets-and-upgrade-kits.html
MWM (Cat Energy Solutions)	TCG 3020 gas engine specs (1,380–2,300 kW, up to 45% efficiency NG)	https://www.mwm.net/en/gas-engines-gensets/gas-engine-tcg-3020-v20/
MWM	TCG 3020 engines/retrofit kits for up to 25 vol% H ₂	https://www.mwm.net/en/news/news-releases/mwm-launches-tcg-3020-gas-engines-and-retrofit-kits-for-hydrogen-operation-with-admixture-of-up-to-25-vol/
MWM	TCG 3016 25 vol% H ₂	https://www.mwm.net/en/news/news-releases/mwm-tcg-3016-gas-engines-and-retrofit-kits-for-hydrogen-operation-with-admixture-of-up-to-25-vol-now-available/
INNIO Jenbacher	J420 specs (up to 1,694 kW, up to 42.6% electrical efficiency)	https://www.jenbacher.com/en/gas-engines/type-4/jenbacher-j420/
INNIO	3 MW 100% H ₂ data-centre backup demonstration (23 Apr 2026)	https://www.innio.com/en/news-media/press-releases/innio-and-net-zero-innovation-hub-deliver-world-first-3-mw-demonstration-of-100-hydrogen-backup-power-for-data-centers/
INNIO	First 100% H ₂ engine power plant in APAC (Hyosung, Ulsan, Korea)	https://www.innio.com/en/news-media/press-releases/innio-technology-selected-for-first-100-hydrogen-engine-power-plant-in-asia-pacific/
Rolls-Royce	TÜV SÜD H ₂ -ready certificate for mtu Series 4000 (25 vol% + 100 vol%)	https://www.rolls-royce.com/media/press-releases/2024/11-7-2024-energy-transition-rr-reaches-milestone-with-h2-ready-certificate-for-mtu-engines.aspx
Rolls-Royce	Fast-start mtu gensets, Series 4000 L64, 2.8 MW from 2026	https://www.rolls-royce.com/media/press-releases/2025/02-10-2025-rr-introduces-fast-start-mtu-gas-gensets-for-powering-data-centers.aspx
2G Energy	H ₂ -ready / agenitor H ₂ models	https://2-g.com/en/solutions/h2-ready
2G Energy	agenitor product range	https://2-g.com/en/products/chp-systems/agenitor
DEUTZ	Hydrogen engines (TCG 7.8 H ₂ , 200 kW)	https://www.deutz.com/en/products/hydrogen-engines/
DEUTZ	H ₂ gensets in operation in China	https://www.deutz.com/en/news/press-releases/news-detail/deutz-puts-hydrogen-powered-generator-sets-into-operation-in-china/
HD Hyundai	Infracore hydrogen engine roadmap (HX12; power-gen target 2026)	https://esg.hd.com/en/news/842
HD Hyundai	1,500 h on a 200 kW H ₂ generator	https://esg.hd.com/en/news/876
Everlence / MAN ES	Hydrogen power solutions (35/44DF H ₂ research engine; 25% H ₂ on 35/44GTS & 51/60G)	https://www.man-es.com/energy-storage/campaigns/hydrogen-power-solutions
Cummins	B6.7H hydrogen ICE development completed (on-highway)	https://www.route-one.net/suppliers/cummins-b6-7h-hydrogen-engine-development-is-completed/

OEM / Org	Item	URL
Liebherr	H966 hydrogen engine (off-highway, no genset product)	https://www.liebherr.com/en-int/components/solutions/combustion-engines/product-portfolio-diesel-engines/h966-6725400
Yanmar	~500 kW pilot-ignition hydrogen 4-stroke at rated output (30 Oct 2024)	https://www.yanmar.com/global/news/2024/10/30/143738.html
Singapore EDB	Hydrogen-ready power plant by 2026 (Keppel Sakra Cogen, 600 MW, 30% H ₂)	https://www.edb.gov.sg/en/business-insights/insights/new-hydrogen-ready-power-plant-to-be-built-by-2026-as-singapore-seeks-greener-energy-generators.html
Enlit World	GE Vernova 9HA.01 order, 600 MW hydrogen-capable CCGT, Pulau Seraya	https://www.enlit.world/library/ge-vernova-confirms-order-for-hydrogen-ready-power-plant-in-singapore
Senoko Energy / Gentari	Hydrogen import from Malaysia into Senoko CCGTs, supply from ~2029	https://www.malaymail.com/amp/news/money/mediaoutreach/2024/10/24/senoko-energy-and-gentari-collaborate-to-enhance-sustainable-energy-solutions-in-singapore-with-hydrogen/335624

Full survey record with dated changelog entry: memory.md, MYSGH2project repository (section 6 and section 7), entry “H₂ reciprocating gas engine (genset) OEM survey”, 2026-09-22.